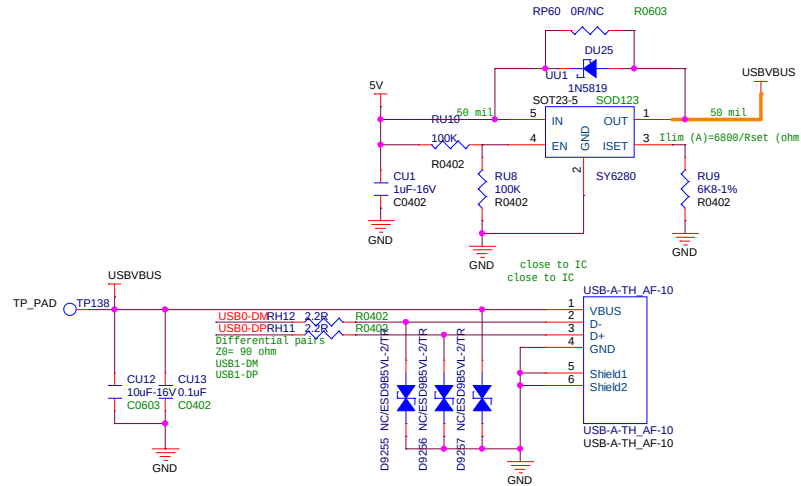
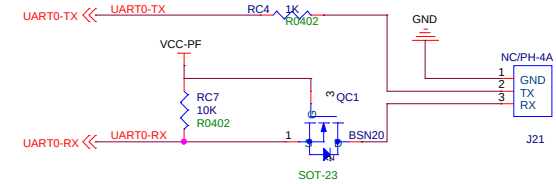


USB

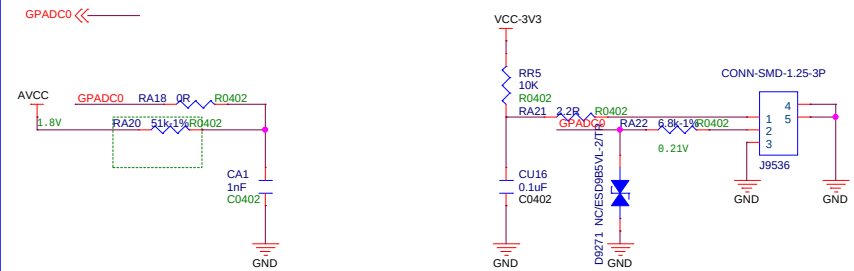
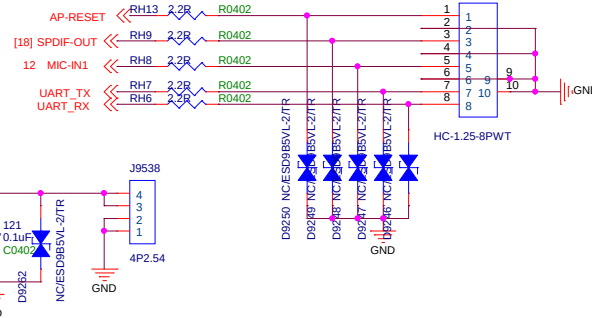
USB0-DM
USB0-DP



UART DEBUG

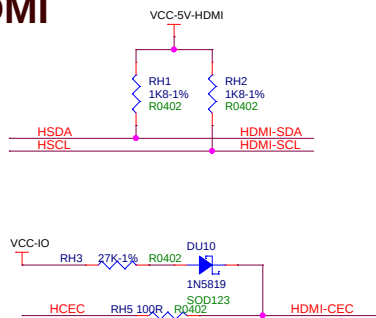


KEY ADC

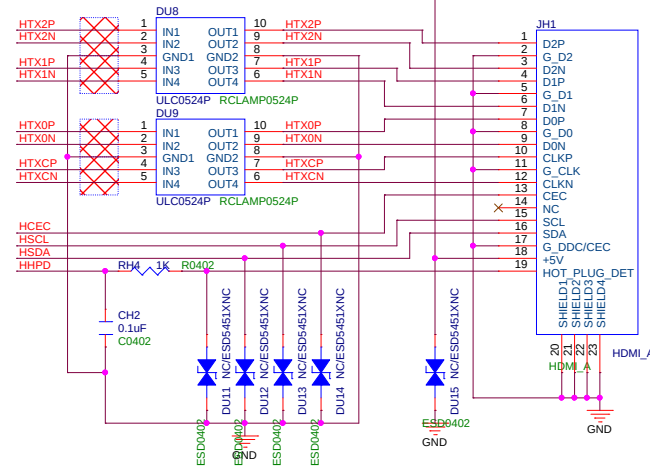


HDMI

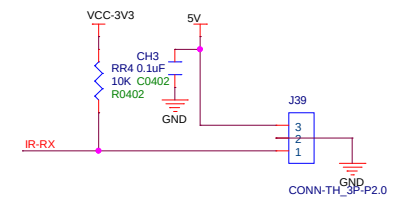
HCEC
HHPD
HSCL
HSDA
HTX0N
HTX0P
HTX1N
HTX1P
HTX2N
HTX2P
HTXCN
HTXCP



NOTE:
Differential pairs
Z0= 100 ohm



IR-RX



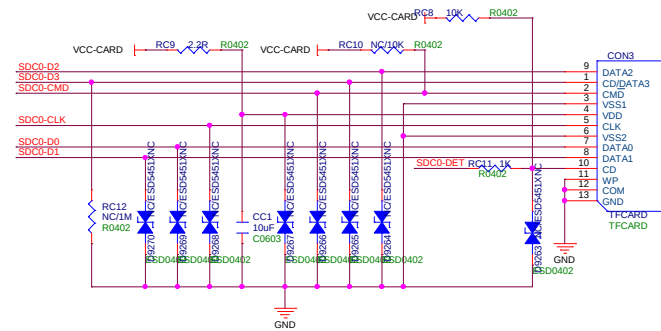
VERSION HISTORY

Index:

- P01 VERSION HISTORY
- P02 BLOCK DIAGRAM
- P03 POWER TREE
- P04 GPIO ASSIGNMENT
- P05 POWER
- P06 SOC1
- P07 SOC2
- P08 DDR3
- P09 FLASH
- P10 AUDIO
- P11 USB CARD HDMI
- P12 WIFI BT
- OPTIONAL
- P13 DDR2
- P14 USB WIFI

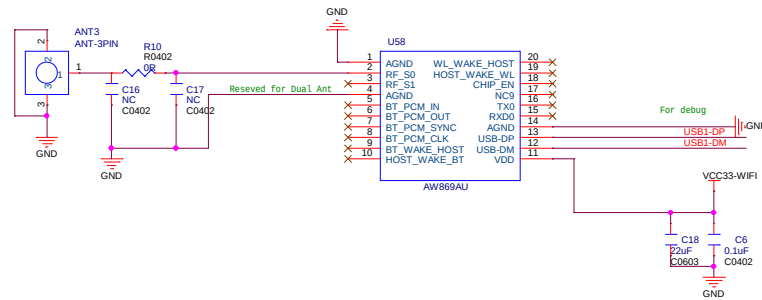
Revision	Description	Date	Drawn	Checked	Approved
Ver 0.1	Releas version	2020-12-31			
Ver 1.0	AddAW859 Add UART2 /5 and SPI test point	2021-04-07	YJY	JIAYONG	YINWEI
Ver 1.1	ADD DDR2 ADD FALE FEL KEY ADD USB WIFI ADD DCDC4 for dram	2021-11-21	WJH	YJY	YINWEI
Ver 1.2	Merge VDD-CPU and vdd-SYS VCC-DRAM change to ext DCDC	2022-01-05	WJH	YJY	YINWEI

SDC0/JTAG Debug

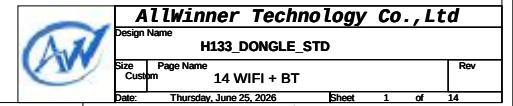


SDC0-D1
SDC0-D0
SDC0-CLK
SDC0-CMD
SDC0-D3
SDC0-D2
SDC0-DET

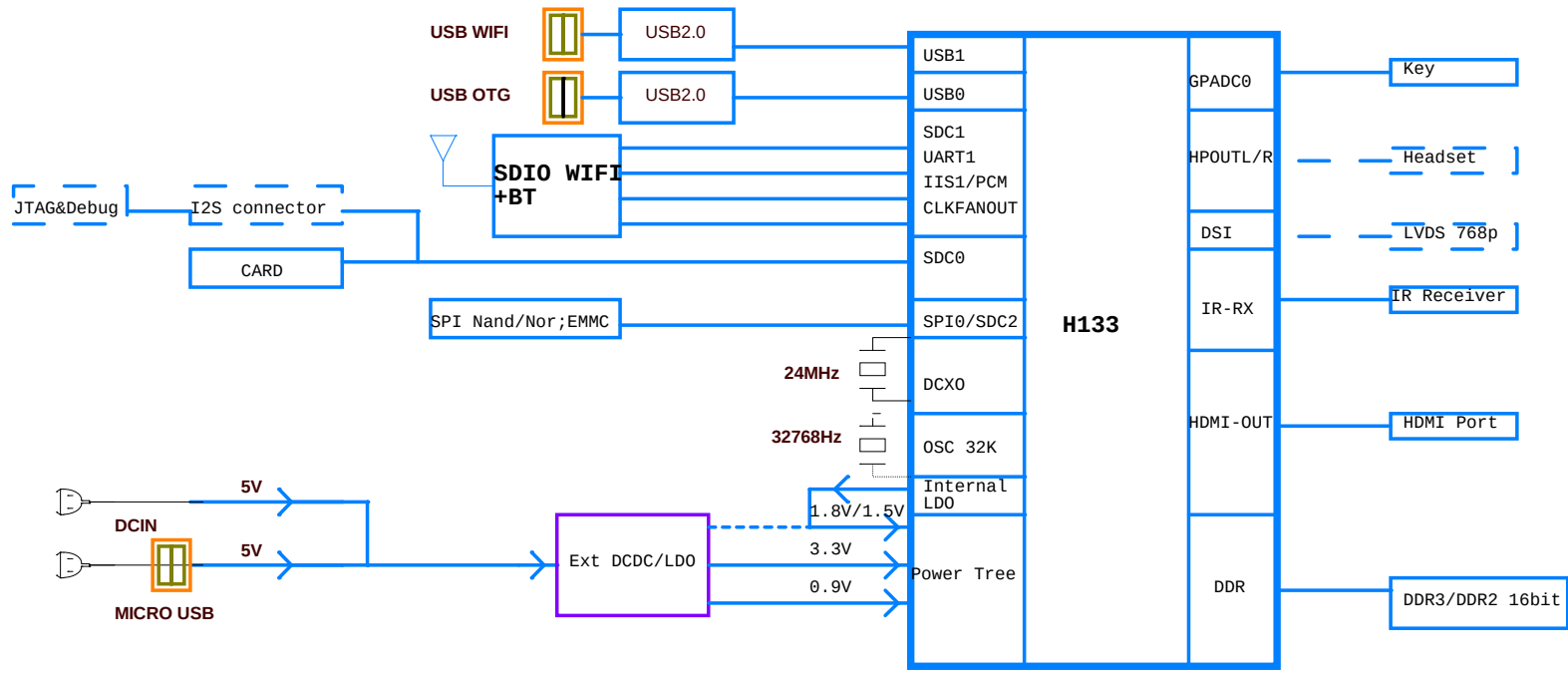
USB WIFI



USB1-DM
USB1-DP

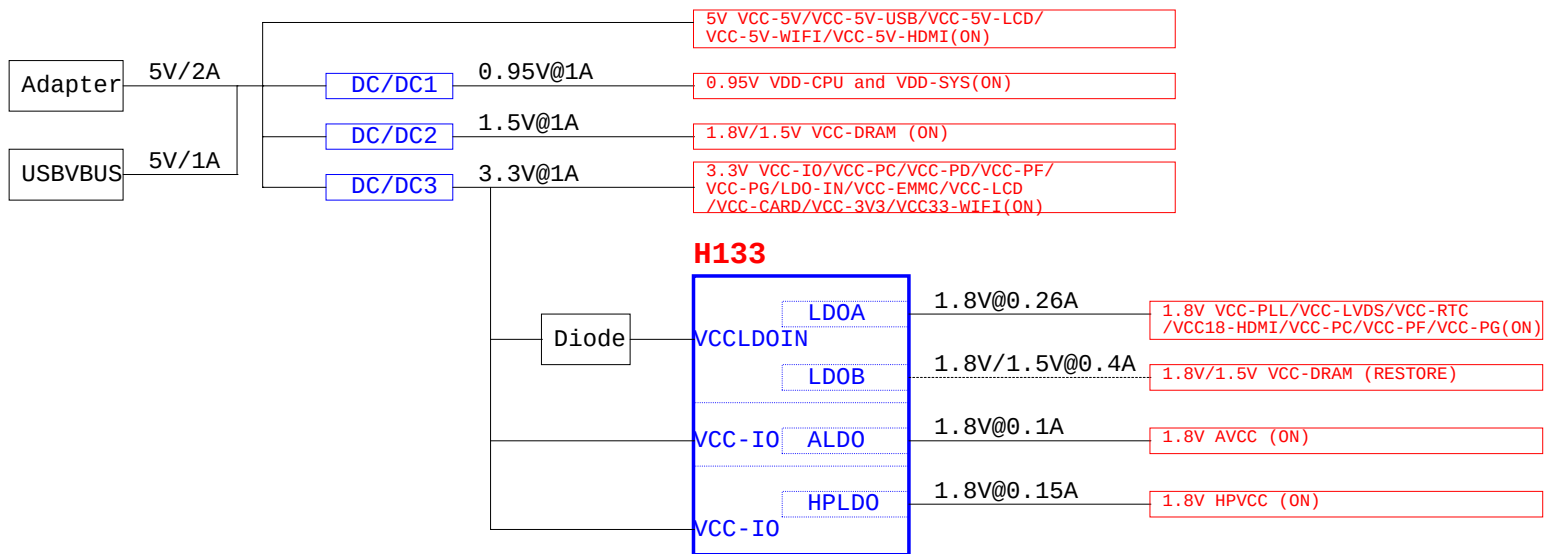


BLOCK



☐ DEFAULT POWER ON

☐ DEFAULT POWER OFF



GPIO ASSIGNMENT

Ball Number	Ball Name	GPIO Multiplex Function
D15	PB0	PWM3/IR_TX/TWI2_SCK/SPI1_WP/DBI_TE/UART0_TX/UART2_TX/SPDIF_OUT/PB_EINT0
D14	PB1	PWM4/I2S2_DOUT3/TWI2_SDA/I2S2_DIN3/UART0_RX/UART2_RX/IR_RX/PB_EINT1
D13	PB8	DMIC_DATA3/PWM5/TWI2_SCK/SPI1_HOLD/DBI_DCX/DBI_WRX/UART0_TX/UART1_TX/PB_EINT8
C14	PB9	DMIC_DATA2/PWM6/TWI2_SDA/SPI1_MISO/DBI_SDI/DBI_TE/DBI_DCX/UART0_RX/UART1_RX/PB_EINT9
C13	PB10	DMIC_DATA1/PWM7/TWI0_SCK/SPI1_MOSI/DBI_SDO/CLK_FANOUT0/UART1_RTS/PB_EINT10
B15	PB11	DMIC_DATA0/PWM2/TWI0_SDA/SPI1_CLK/DBI_SCLK/CLK_FANOUT1/UART1_CTS/PB_EINT11
B14	PB12	DMIC_CLK/PWM0/SPDIF_IN/SPI1_CS/DBI_CSX/CLK_FANOUT2/IR_RX/PB_EINT12

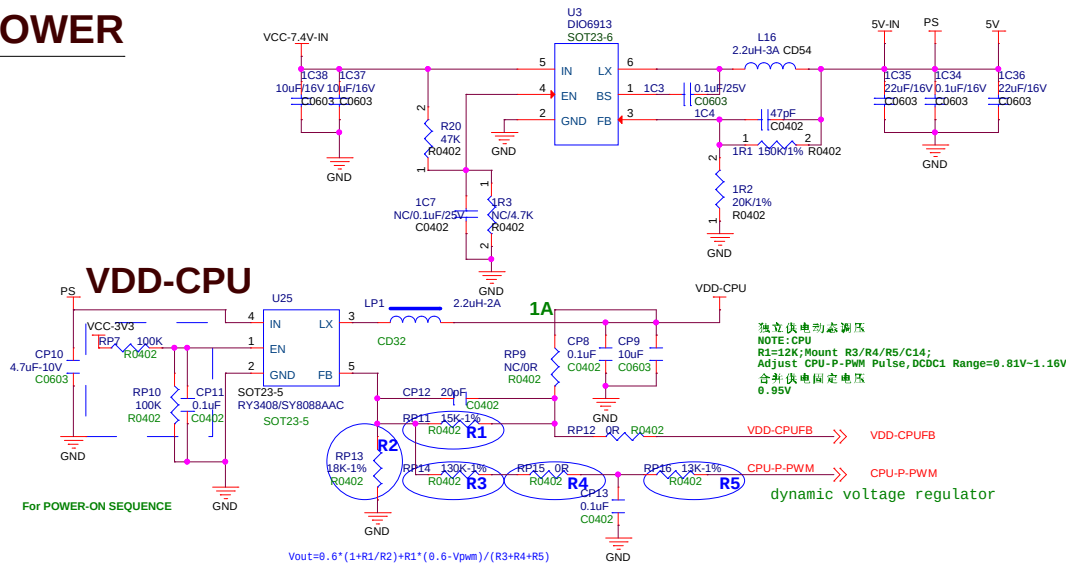
Ball Number	Ball Name	GPIO Multiplex Function
F3	PC2	SPI0_CLK/SDC2_CLK/PC_EINT2
F2	PC3	SPI0_CS0/SDC2_CMD/PC_EINT3
F1	PC4	SPI0_MOSI/SDC2_D2/B00T_SEL0/PC_EINT4
G3	PC5	SPI0_MISO/SDC2_D1/B00T_SEL1/PC_EINT5
G2	PC6	SPI0_WP/SDC2_D0/UART3_TX/TWI3_SCK/DBG_CLK/PC_EINT6
H3	PC7	SPI0_HOLD/SDC2_D3/UART3_RX/TWI3_SDA/TCON_TRIG/PC_EINT7

Ball Number	Ball Name	GPIO Multiplex Function
N15	PD0	LCD0_D2/LVDS0_V0P/DSI_D0P/TWI0_SCK/PD_EINT0
N14	PD1	LCD0_D3/LVDS0_V0N/DSI_D0N/UART2_TX/PD_EINT1
M15	PD2	LCD0_D4/LVDS0_V1P/DSI_D1P/UART2_RX/PD_EINT2
M14	PD3	LCD0_D5/LVDS0_V1N/DSI_D1N/UART2_RTS/PD_EINT3
L15	PD4	LCD0_D6/LVDS0_V2P/DSI_CKP/UART2_CTS/PD_EINT4
L14	PD5	LCD0_D7/LVDS0_V2N/DSI_CKN/UART5_TX/PD_EINT5
K15	PD6	LCD0_D10/LVDS0_CKP/DSI_D2P/UART5_RX/PD_EINT6
K14	PD7	LCD0_D11/LVDS0_CKN/DSI_D2N/UART4_TX/PD_EINT7
J15	PD8	LCD0_D12/LVDS0_V3P/DSI_D3P/UART4_RX/PD_EINT8
J14	PD9	LCD0_D13/LVDS0_V3N/DSI_D3N/PWM6/PD_EINT9

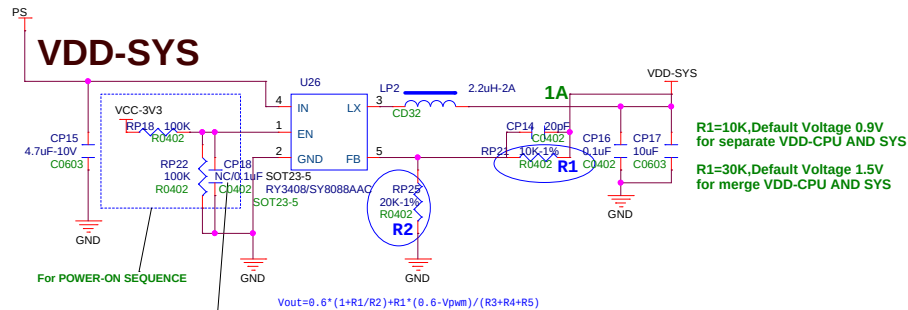
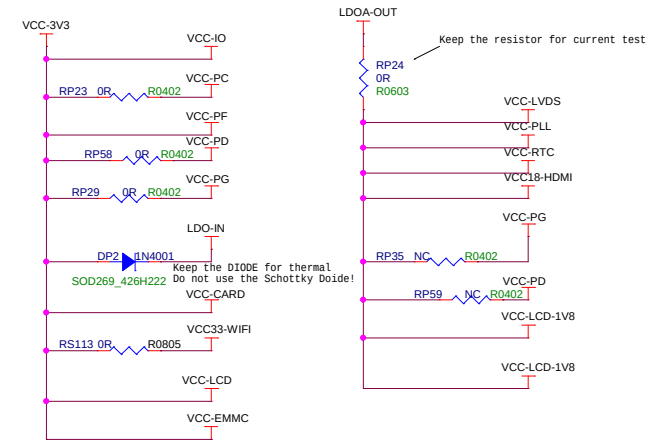
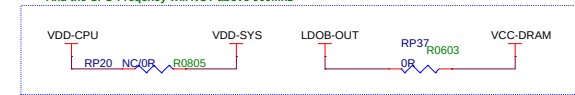
Ball Number	Ball Name	GPIO Multiplex Function
B8	PG0	SDC1_CLK/UART3_TX/RGMII_RXCTRL/RMII_CRS_DV/PWM7/PG_EINT0
C9	PG1	SDC1_CMD/UART3_RX/RGMII_RXD0/RMII_RXD0/PWM6/PG_EINT1
A8	PG2	SDC1_D0/UART3_RTS/RGMII_RXD1/RMII_RXD1/UART4_TX/PG_EINT2
B7	PG3	SDC1_D1/UART3_CTS/RGMII_TXCK/RMII_TXCK/UART4_RX/PG_EINT3
A6	PG4	SDC1_D2/UART5_TX/RGMII_TXD0/RMII_TXD0/PWM5/PG_EINT4
C7	PG5	SDC1_D3/UART5_RX/RGMII_TXD1/RMII_TXD1/PWM4/PG_EINT5
B4	PG6	UART1_TX/TWI2_SCK/RGMII_TXD2/PWM1/PG_EINT6
A3	PG7	UART1_RX/TWI2_SDA/RGMII_TXD3/SPDIF_IN/PG_EINT7
B3	PG8	UART1_RTS/TWI1_SCK/RGMII_RXD2/UART3_TX/PG_EINT8
A2	PG9	UART1_CTS/TWI1_SDA/RGMII_RXD3/UART3_RX/PG_EINT9
C4	PG10	PWM3/TWI3_SCK/RGMII_RXCK/CLK_FANOUT0/IR_RX/PG_EINT10
B6	PG11	I2S1_MCLK/TWI3_SDA/EPHY_25M/CLK_FANOUT1/TCON_TRIG/PG_EINT11
C6	PG12	I2S1_LRCK/TWI0_SCK/RGMII_TXCTRL/RMII_TXEN/CLK_FANOUT2/PWM0/UART1_TX/PG_EINT12
B5	PG13	I2S1_BCLK/TWI0_SDA/RGMII_CLKIN/RMII_RXER/PWM2/LEDC_D0/UART1_RX/PG_EINT13
C5	PG14	I2S1_DIN0/TWI2_SCK/MDC/I2S1_DOUT1/SPI0_WP/UART1_RTS/PG_EINT14
A4	PG15	I2S1_DOUT0/TWI2_SDA/MDIO/I2S1_DIN1/SPI0_HOLD/UART1_CTS/PG_EINT15
B2	PG16	IR_RX/TCON_TRIG/PWM5/CLK_FANOUT2/SPDIF_IN/LEDC_D0/PG_EINT16
C10	PG17	UART2_TX/TWI3_SCK/PWM7/CLK_FANOUT0/IR_TX/UART0_TX/PG_EINT17
B9	PG18	UART2_RX/TWI3_SDA/PWM6/CLK_FANOUT1/SPDIF_OUT/UART0_RX/PG_EINT18

Ball Number	Ball Name	GPIO Multiplex Function
B1	PF0	SDC0_D1/JTAG_MS/R_JTAG_MS/I2S2_DOUT1/I2S2_DIN0/PF_EINT0
C3	PF1	SDC0_D0/JTAG_DI/R_JTAG_DI/I2S2_DOUT0/I2S2_DIN1/PF_EINT1
C2	PF2	SDC0_CLK/UART0_TX/TWI0_SCK/LEDC_D0/SPDIF_IN/PF_EINT2
D3	PF3	SDC0_CMD/JTAG_D0/R_JTAG_D0/I2S2_BCLK/PF_EINT3
D2	PF4	SDC0_D3/UART0_RX/TWI0_SDA/PWM6/IR_TX/PF_EINT4
D1	PF5	SDC0_D2/JTAG_CK/R_JTAG_CK/I2S2_LRCK/PF_EINT5
E2	PF6	SPDIF_OUT/IR_RX/I2S2_MCLK/PWM5/PF_EINT6

POWER

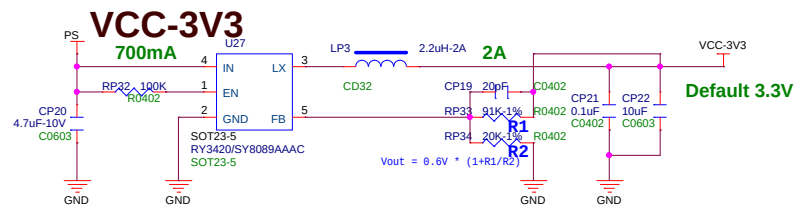
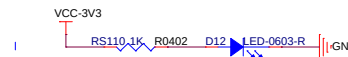


If merge VDD-CPU and VDD-SYS
The both voltage must be fixed at 0.95V
And the CPU-Frequency will NOT above 900Mhz



此DCDC给VCC-DRAM供电，CP18要NC，才能满足时序要求

POWER-LED



	DCDC1 (EXT)	DCDC2 (EXT)	DCDC3 (EXT)	DCDC4 (EXT)	LDOB (IN)	LDOA (IN)	REMARK
Default PLAN	VDD-CPU VDD-SYS (0.95V)	VCC-DRAM 1.5/1.8V	VCC-WIFI VCC-3V3 LDOIN	NC	NC	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.The Power consume is below2.5W@4K30,And the thermal will be very low.
Performance PLAN	VDD-CPU (0.8-1.16V)	VDD-SYS 0.9V	VCC-WIFI VCC-3V3 LDOIN	VCC-DRAM 1.5/1.8V	NC	VCC-PLL VCC-RTC VCC-1V8	Separate VDD-CPU and VDD-SYS ,The Cpu-frequency upon to 1.2Ghz .The Power consume is below2.5W@4K30,And the thermal will be very low.
Econormy PLAN	VDD-CPU VDD-SYS (0.95V)	NC	VCC-WIFI VCC-3V3 LDOIN	NC	VCC-DRAM 1.5/1.8V	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.VCC-Dram be supplied by internal LDOB,instead of Ext DCDC.



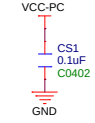
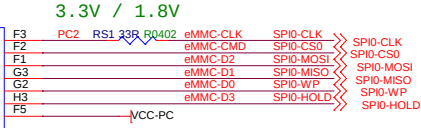
Allwinner Technology Co., Ltd			
Design Name		H133_DONGLE_STD	
Size	Page Name	Rev	
A3	05 POWER		
Date:	Thursday, June 25, 2026	Sheet	5 of 12

SOC1

U11C

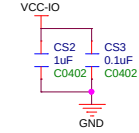
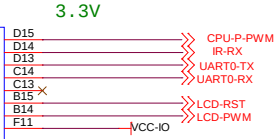
PC

PC2/SPI0_CLK/SDC2_CLK/PC_EINT2
PC3/SPI0_CS0/SDC2_CMD/PC_EINT3
PC4/SPI0_MOSI/SDC2_D2/BOOT_SEL0/PC_EINT4
PC5/SPI0_MISO/SDC2_D1/BOOT_SEL1/PC_EINT5
PC6/SPI0_WP/SDC2_D0/UART3_TX/TWI3_SCK/DBG_CLK/PC_EINT6
PC7/SPI0_HOLD/SDC2_D3/UART3_RX/TWI3_SDA/TCON_TRIG/PC_EINT7
VCC_PC



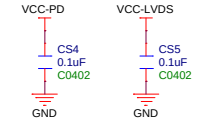
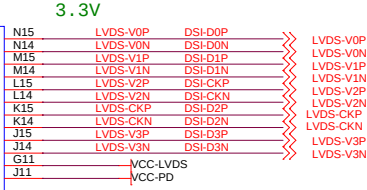
PB

PB0/PWM3/IR_TX/TWI2_SCK/SPI1_WP/DBI_TE/UART0_TX/UART2_TX/SPDIF_OUT/PB_EINT0
PB1/PWM4/I2S2_DOUT3/TWI2_SDA/I2S2_DIN3/UART0_RX/UART2_RX/IR_RX/PB_EINT1
PB8/DMIC_DATA3/PWM5/TWI2_SCK/SPI1_HOLD/DBI_DCX/DBI_WRX/UART0_TX/UART1_TX/PB_EINT8
PB9/DMIC_DATA2/PWM6/TWI2_SDA/SPI1_MISO/DBI_SDI/DBI_TE/DBI_DCX/UART0_RX/UART1_RX/PB_EINT9
PB10/DMIC_DATA1/PWM7/TWI0_SCK/SPI1_MOSI/DBI_SDO/CLK_FANOUT0/UART1_RTS/PB_EINT10
PB11/DMIC_DATA0/PWM2/TWI0_SDA/SPI1_CLK/DBI_SCLK/CLK_FANOUT1/UART1_CTS/PB_EINT11
PB12/DMIC_CLK/PWM0/SPDIF_IN/SPI1_CS/DBI_CSX/CLK_FANOUT2/IR_RX/PB_EINT12
VCC_IO



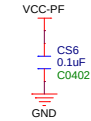
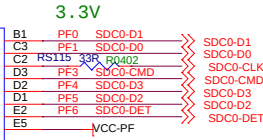
PD

PD0/LCD0_D2/LVDS0_V0P/DSI_D0P/TWI0_SCK/PD_EINT0
PD1/LCD0_D3/LVDS0_V0N/DSI_D0N/UART2_TX/PD_EINT1
PD2/LCD0_D4/LVDS0_V1P/DSI_D1P/UART2_RX/PD_EINT2
PD3/LCD0_D5/LVDS0_V1N/DSI_D1N/UART2_RTS/PD_EINT3
PD4/LCD0_D6/LVDS0_V2P/DSI_D2P/UART2_CTS/PD_EINT4
PD5/LCD0_D7/LVDS0_V2N/DSI_D2N/UART5_TX/PD_EINT5
PD6/LCD0_D10/LVDS0_CK/PD_SCK/PD_SCK/PD_SCK/PD_SCK/PD_EINT6
PD7/LCD0_D11/LVDS0_CKN/DSI_D2N/UART4_TX/PD_EINT7
PD8/LCD0_D12/LVDS0_V3P/DSI_D3P/UART4_RX/PD_EINT8
PD9/LCD0_D13/LVDS0_V3N/DSI_D3N/PWM6/PD_EINT9
VCC_LVDS
VCC_PD



PF

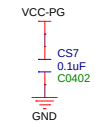
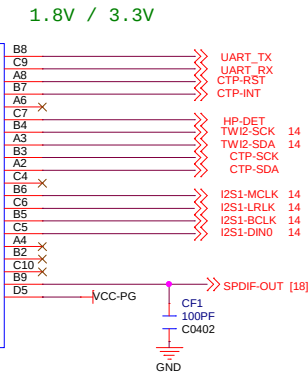
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PF1/SDC0_D0/JTAG_DI/R_JTAG_DI/I2S2_DOUT0/I2S2_DIN1/PF_EINT1
PF2/SDC0_CLK/UART0_TX/TWI0_SCK/LEDC_DO/SPDIF_IN/PF_EINT2
PF3/SDC0_CMD/JTAG_DO/R_JTAG_DO/I2S2_BCLK/PF_EINT3
PF4/SDC0_D3/UART0_RX/TWI0_SDA/PWM6/IR_TX/PF_EINT4
PF5/SDC0_D2/JTAG_CK/R_JTAG_CK/I2S2_LRCK/PF_EINT5
PF6/SPDIF_OUT/IR_RX/I2S2_MCLK/PWM5/PF_EINT6
VCC_PF



U11G

PG

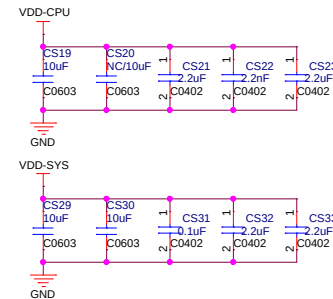
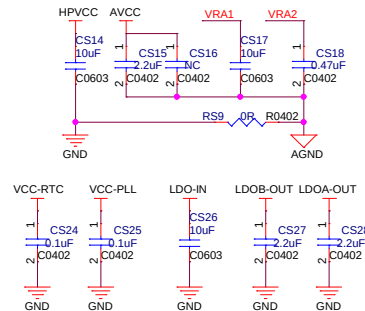
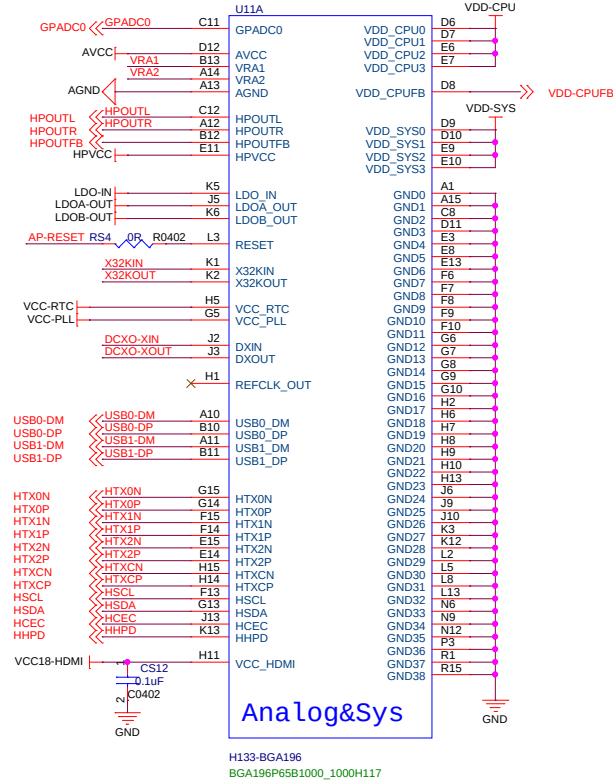
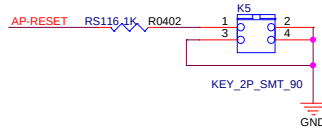
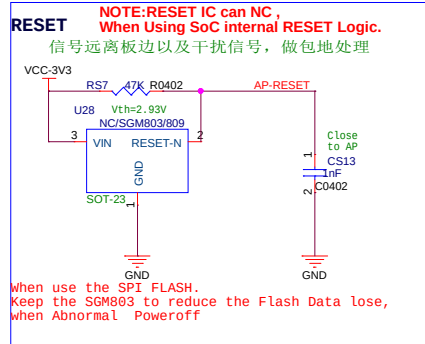
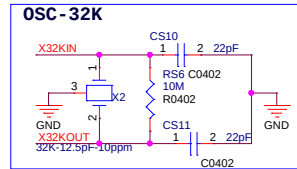
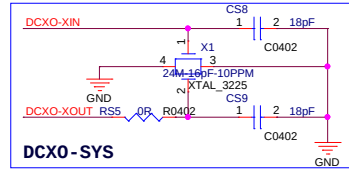
PG0/SDC1_CLK/UART3_TX/RGMII_RXCTRL/RMII_CRS_DV/PWM7/PG_EINT0
PG1/SDC1_CMD/UART3_RX/RGMII_RXD0/RMII_RXD0/PWM6/PG_EINT1
PG2/SDC1_D0/UART3_RTS/RGMII_RXD1/RMII_RXD1/UART4_TX/PG_EINT2
PG3/SDC1_D1/UART3_CTS/RGMII_TXCK/RMII_TXCK/UART4_RX/PG_EINT3
PG4/SDC1_D2/UART5_TX/RGMII_TXD0/RMII_TXD0/PWM5/PG_EINT4
PG5/SDC1_D3/UART5_RX/RGMII_TXD1/RMII_TXD1/PWM4/PG_EINT5
PG6/UART1_TX/TWI2_SCK/RGMII_TXD2/PWM1/PG_EINT6
PG7/UART1_RX/TWI2_SDA/RGMII_TXD3/SPDIF_IN/PG_EINT7
PG8/UART1_RTS/TWI2_SCK/RGMII_RXD2/UART3_TX/PG_EINT8
PG9/UART1_CTS/TWI1_SDA/RGMII_RXD3/UART3_RX/PG_EINT9
PG10/PWM3/TWI3_SCK/RGMII_RXCK/CLK_FANOUT0/IR_RX/PG_EINT10
PG11/I2S1_MCLK/TWI3_SDA/EPHY_25MCLK_FANOUT1/TCON_TRIG/PG_EINT11
PG12/I2S1_LRCK/TWI0_SCK/RGMII_TXCTRL/RMII_TXENCLK_FANOUT2/PWM0/UART1_TX/PG_EINT12
PG13/I2S1_BCLK/TWI0_SDA/RGMII_CLK/RMII_RXER/PWM2/LEDC_DO/UART1_RX/PG_EINT13
PG14/I2S1_DIN0/TWI2_SCK/MDC/I2S1_DOUT1/SPI0_WP/UART1_RTS/PG_EINT14
PG15/I2S1_DOUT0/TWI2_SDA/MDC/I2S1_DIN1/SPI0_HOLD/UART1_CTS/PG_EINT15
PG16/IR_RX/TCON_TRIG/PWM5/CLK_FANOUT2/SPDIF_IN/LEDC_DO/PG_EINT16
PG17/UART2_TX/TWI3_SCK/PWM7/CLK_FANOUT0/IR_TX/UART0_TX/PG_EINT17
PG18/UART2_RX/TWI3_SDA/PWM6/CLK_FANOUT1/SPDIF_OUT/UART0_RX/PG_EINT18
VCC_PG

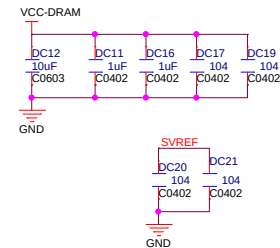
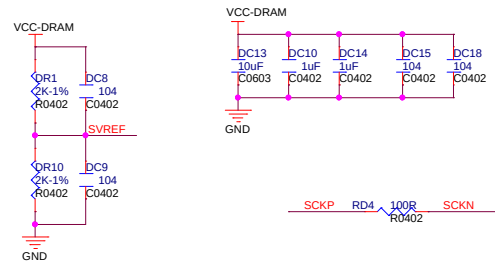
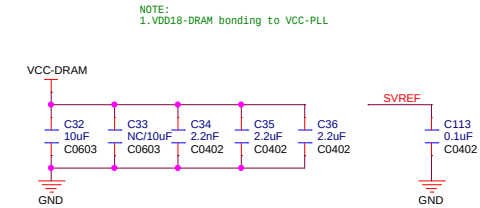


H133-BGA196

BGA196P6581000_1000H117

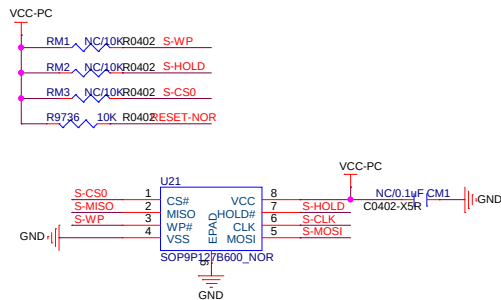
SOC2



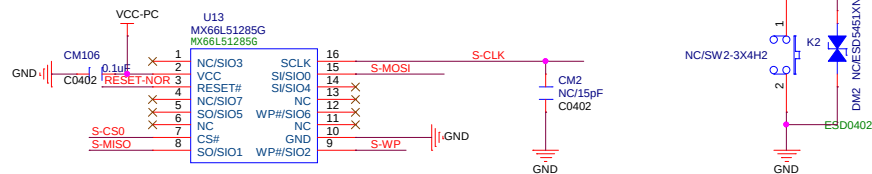
[illegible]

SPI

SPI0-CLK << S-CLK
 SPI0-CS0 << S-CS0
 SPI0-MOSI << S-MOSI
 SPI0-MISO << S-MISO
 SPI0-WP << S-WP
 SPI0-HOLD << S-HOLD



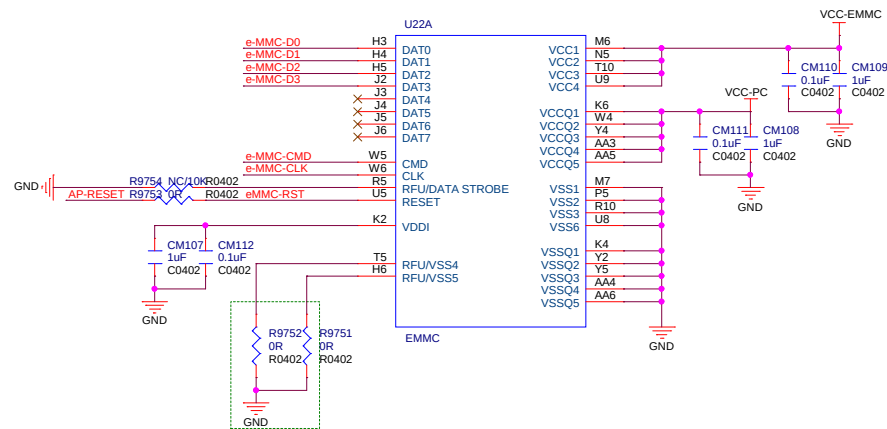
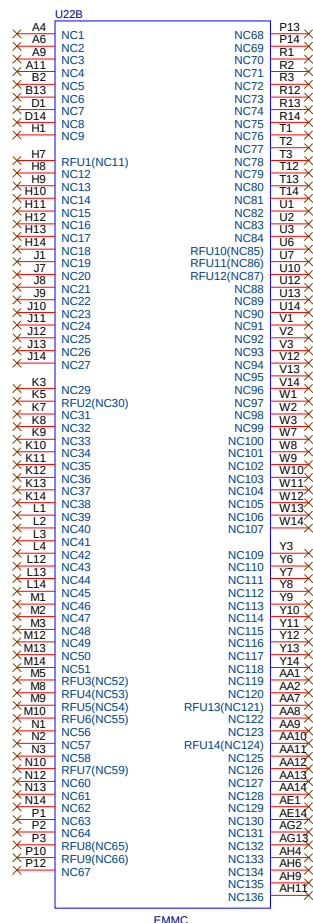
UBOOT



EMMC

SPI FLASH and EMMC only can choose one of them.
 Both Layout in PCB will effect the SI

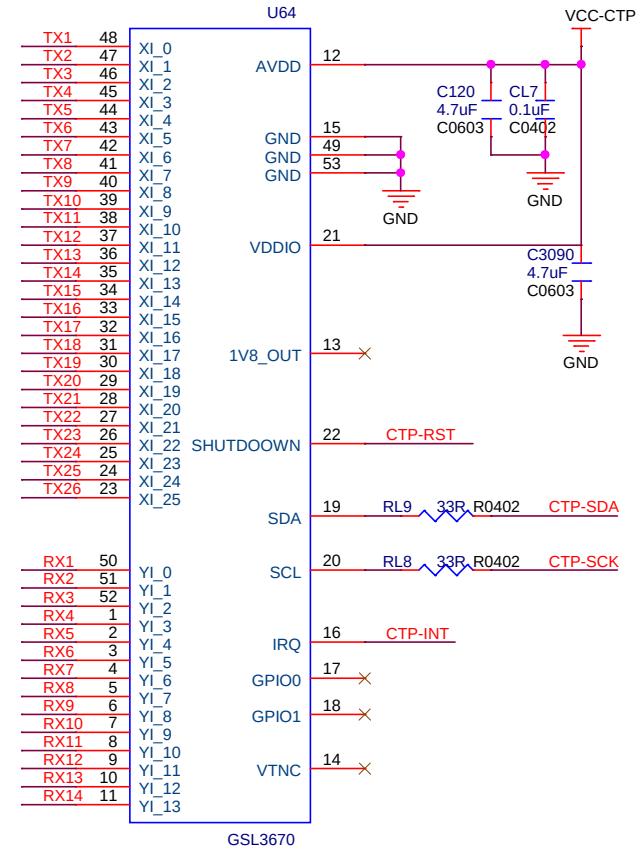
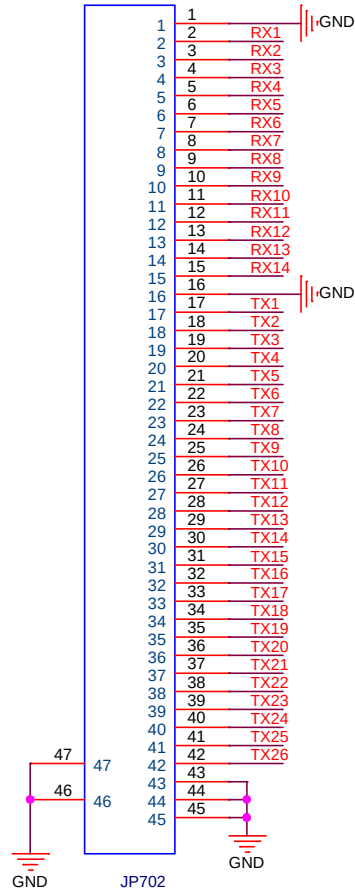
SPI0-CLK << R9748 0R/NB0402 eMMC-CLK
 SPI0-CS0 << R9746 0R/NB0402 eMMC-CMD
 SPI0-MOSI << R9747 0R/NB0402 eMMC-D2
 SPI0-MISO << R9748 0R/NB0402 eMMC-D1
 SPI0-WP << R9749 0R/NB0402 eMMC-D0
 SPI0-HOLD << R9750 0R/NB0402 eMMC-D3
 AP-RESET << AP-RESET



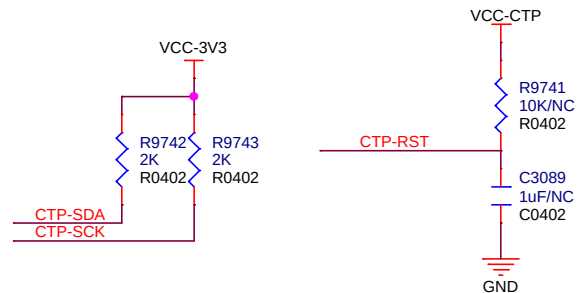
If eMMC is not v5.0/v5.1, then NC this two resistors.

CTP

FFC-SMD_45P-P0.50



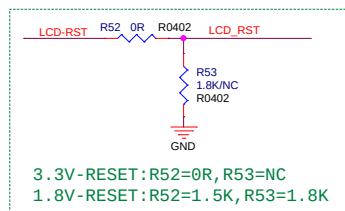
7 CTP-INT
7 CTP-RST
7 CTP-SCK
7 CTP-SDA



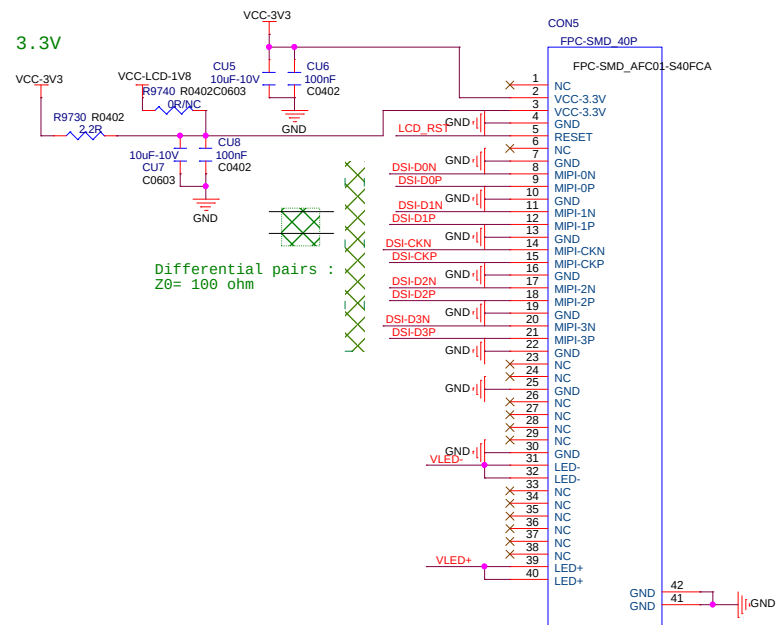
AllWinner Technology Co., Ltd

Design Name		
F133_STD_V1_0		
Size	Page Name	Rev
A4	09 LCD	
Date:	Thursday, June 25, 2026	Sheet 12 of 21

Figure 1 illustrates the DSI interface between the LVDS-V0P/V0N and LVDS-V3P/V3N pairs. The diagram shows the signal paths for LVDS-V0P/V0N, LVDS-V1P/V1N, LVDS-V2P/V2N, and LVDS-V3P/V3N. Each pair is connected to a corresponding DSI signal (DSI-D0P, DSI-D0N, DSI-D1P, DSI-D1N, DSI-CKP, DSI-CKN, DSI-D2P, DSI-D2N, DSI-D3P, DSI-D3N) via a 2.2R R0402 resistor. The LVDS-V0P/V0N pair is connected to DSI-D0P and DSI-D0N. The LVDS-V1P/V1N pair is connected to DSI-D1P and DSI-D1N. The LVDS-V2P/V2N pair is connected to DSI-CKP and DSI-CKN. The LVDS-V3P/V3N pair is connected to DSI-D2P and DSI-D2N. The LVDS-V3P/V3N pair is also connected to DSI-D3P and DSI-D3N via a 2.2R R0402 resistor.

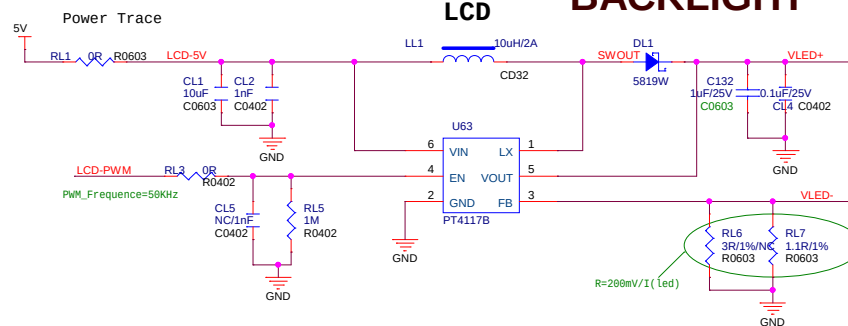


VCC-LCD
IMAX=0.2A



BACKLIGHT

BACKLIGHT

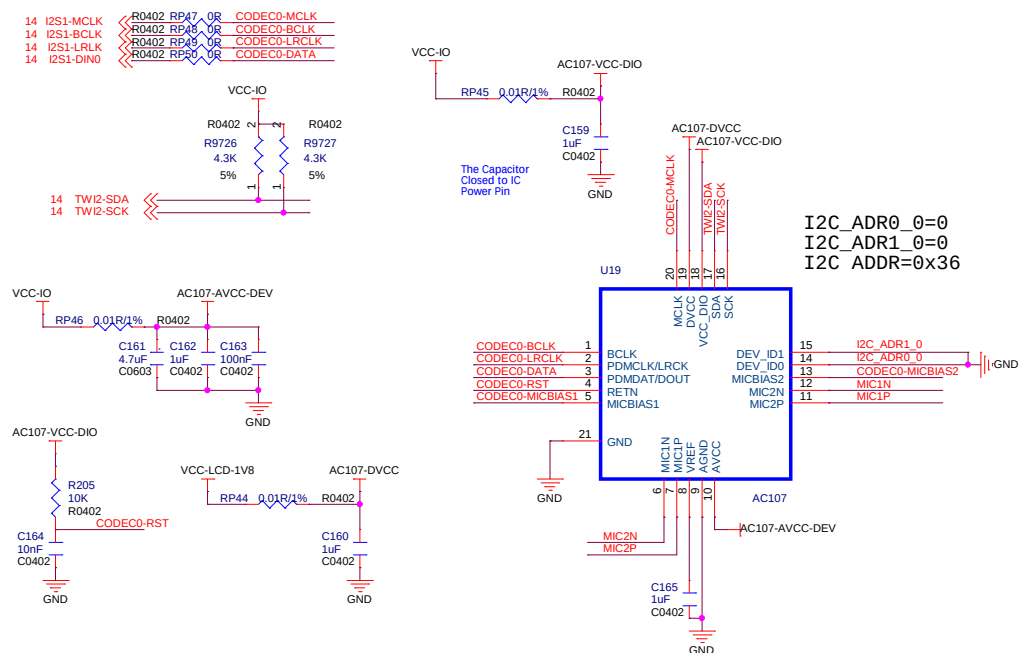


Design Name **A100_A133_B3_AXP717_LPDDR4**

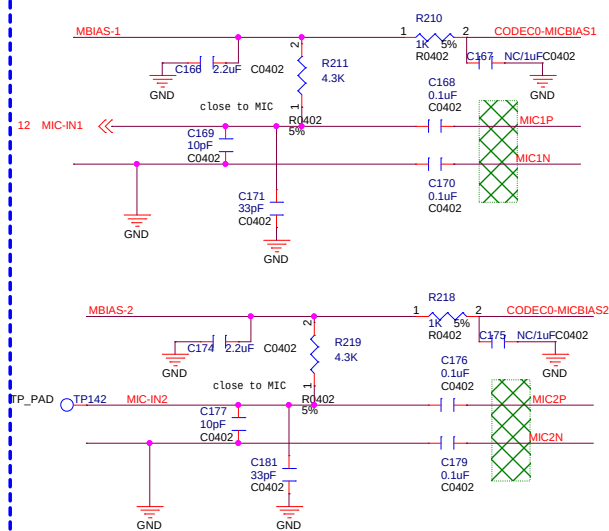
Size	Page Name	Rev
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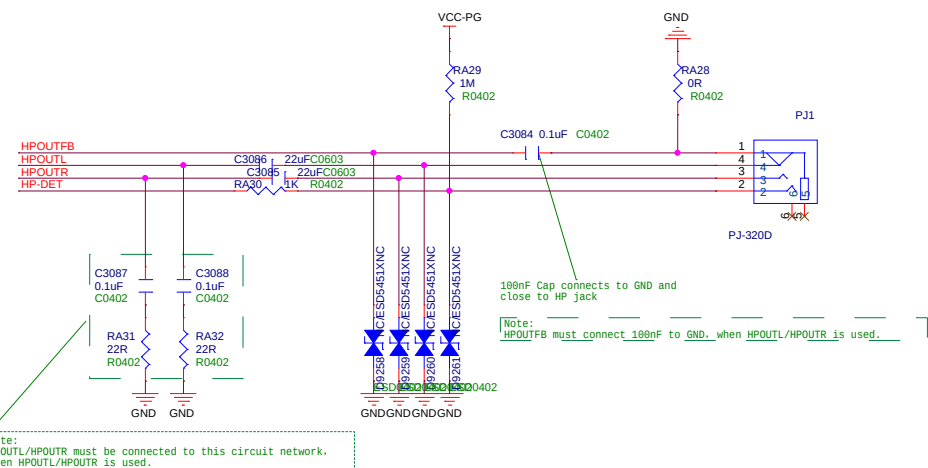
AC107



MIC IN



Audio



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Size A3	Document Number <Doc>	Rev <Rev>	Code <Rev Code>
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